Django Product Management System

Lab Practical Guide & Setup Instructions

PHASE 1: ENVIRONMENT & PROJECT SETUP

Step 1
Create your roll-number folder on Desktop.

Step 2
Open that folder in VS Code / PowerShell.

Step 3
Create Django project:

```
python -m django startproject ProductManagementSystem .
```

Step 4
Create application:

```
python manage.py startapp products
```

PHASE 2: DATA MODEL & APPLICATION CONFIGURATION

Step 5
Open **products/models.py** and add the Product model:

```
from django.db import models

class Product(models.Model):
    name = models.CharField(max_length=100)
    category = models.CharField(max_length=100)
    price = models.DecimalField(max_digits=10, decimal_places=2)
    quantity = models.IntegerField()

    def __str__(self):
        return self.name
```

Step 6
Open **ProductManagementSystem/settings.py** :
Add **'products',** inside the **INSTALLED_APPS** list.

Step 7
Open **products/admin.py** and register the model:

```
from django.contrib import admin
from .models import Product

admin.site.register(Product)
```

PHASE 3: DATABASE MIGRATIONS & SUPERUSER

Step 8
Create migration files:

```
python manage.py makemigrations
```

Step 9
Apply migrations:

```
python manage.py migrate
```

Step 10 & 11
Create administrative superuser account:

```
python manage.py createsuperuser
```

Prompt inputs required:

- **Username:** enter your username
- **Email:** enter your email
- **Password:** enter your password

PHASE 4: RUNNING SERVER & ADMIN OPERATIONS

Step 12 & 13
Start development server and check home URL:

```
python manage.py runserver
```

> Browser-ൽ open ചെയ്യുക: **http://127.0.0.1:8000/**

Step 14 & 15
Access Admin Panel:

> Admin page open ചെയ്യാൻ: **http://127.0.0.1:8000/admin/**

Log in using the same Username and Password created in Step 11.

Step 16, 17, 18 & 19
Adding Product Data:

> 16. Login ചെയ്ത ശേഷം: **Products → Add Product**
> 17. Enter details: **Product Name, Category, Price, Quantity**
> 18. Save ചെയ്യുക.
> 19. **At least 5 products add ചെയ്യുക.**

Step 20
Edit / Delete Products:

> Admin → Products
> അവിടെ നിന്ന് product select ചെയ്ത് **change / delete** ചെയ്യാം.